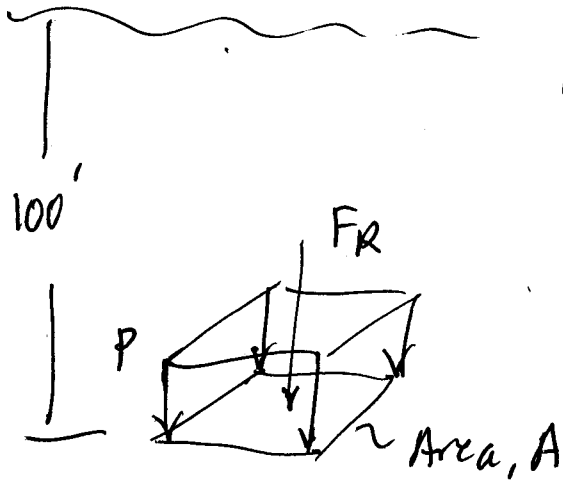


Pressure or Force on a Horiz Plate



$$P = \gamma \cdot h$$

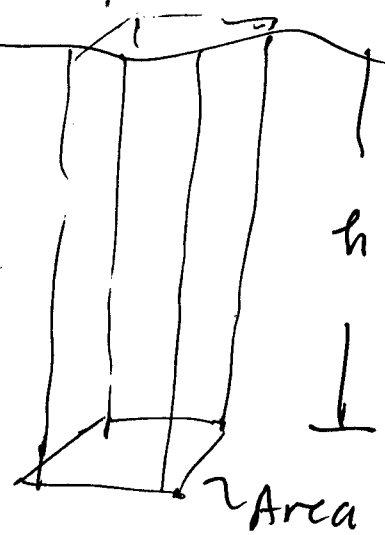
$$F_R = P \cdot \text{Area}$$

$$F_R = (\gamma \cdot h) (\text{Area})$$

But $h \cdot \text{Area} = \text{Volume}$ of Fluid
above the plate

AND, $\gamma \cdot h \cdot A = \text{Weight}$ of Fluid
above the plate

=

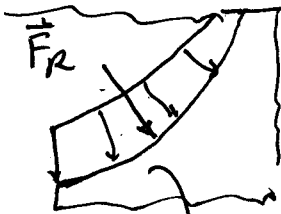


Weight of Column
of Fluid

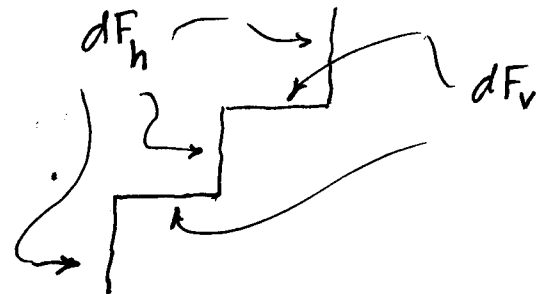
$$F_R = \gamma \cdot h \cdot A$$

$\vec{F}_R = \vec{F}_V + \vec{F}_H$ on a Curved Surface --

How find $\vec{F}_R$?



Represent curve
by discrete steps..



On all horizontal surfaces

$$F_V = \int dF_V = \text{Sum of Wt!}$$

On all vertical surfaces,

$$F = \int dF$$